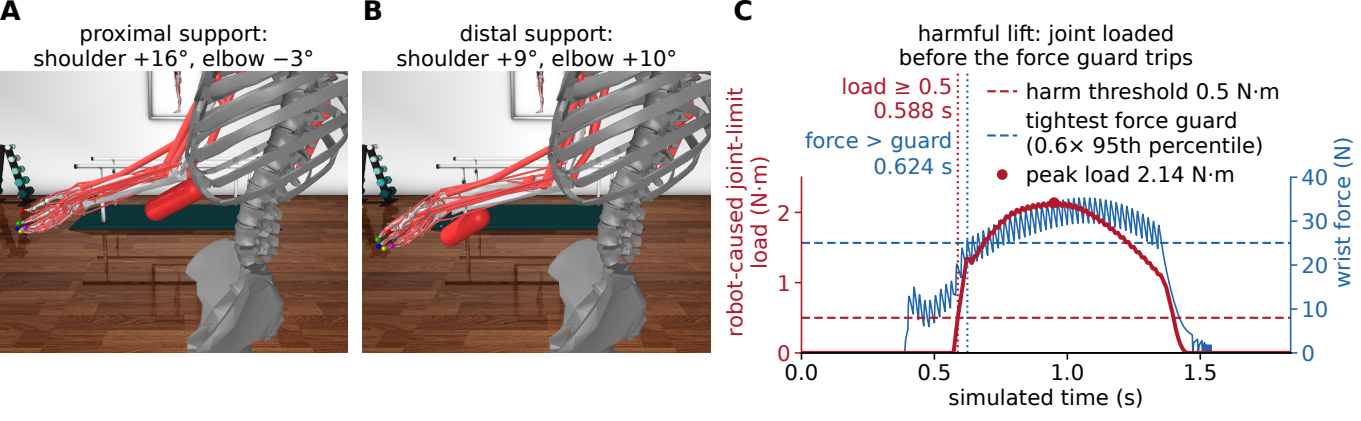

A
proximal support:
shoulder +16°, elbow −3°
B
distal support:
distal support:
shoulder +9°, elbow +10°
C
harmful lift: joint loaded
before the force guard trips
load ≥ 0.5
0.588 s
force > guard
0.624 s
harm threshold 0.5 N·m
tightest force guard
(0.6× 95th percentile)
peak load 2.14 N·m
robot-caused joint-limit
load (N·m)
wrist force (N)
simulated time (s)
0
1
2
0.0
0.5
1.0
1.5
0
10
20
30
40